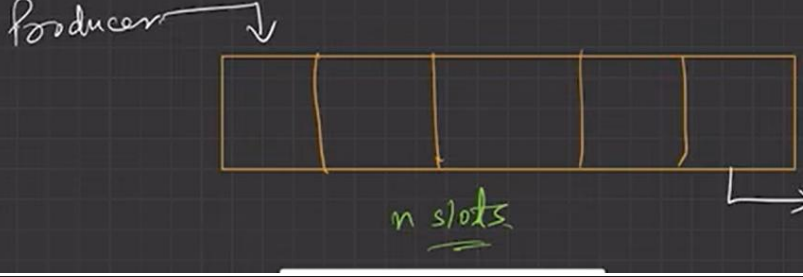
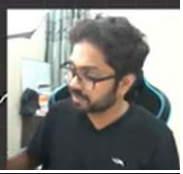
Lec-18

* Producer – Consumer Problem (Bonded Buffer Problem)

1. Producer thread
2. Consumer thread





Problem???

→ C.S (Buffer) → Sync. b/w Producer & consumer.

→ Producer must not insert data when the buffer is full.

→ Consumer must not pick/remove data when the buffer is empty.

Lecture 16: Critical Section Problem || Race condition in OS

Soln Semaphores

① m, mutex → Binary sema. used to acquire lock on buffer.

② empty → a counting sem.
initial value is n
↳ tracks empty slots.

③ full → tracks filled slots.
— initial = 0.



Producer

```
do {
    wait(empty); // wait until
                 empty > 0
                 then, empty → value --
    wait(mutex);
    // C.S, add data to buffer
    signal(mutex);
    signal(full); // increment full
                  → value)
} while(1)
```

Consumer

```
do {
    wait(full); // wait until
                full > 0, then full --;
    wait(mutex);
    // remove data from Buffer
    signal(mutex);
    signal(empty); // increment
                   empty
} while(1)
```

